Graphs and characteristics of quadratic functions

1

a Maximum

b 25

2

a $x = -10$

b $m = -697$

3

a The function has a minimum value.

b 4

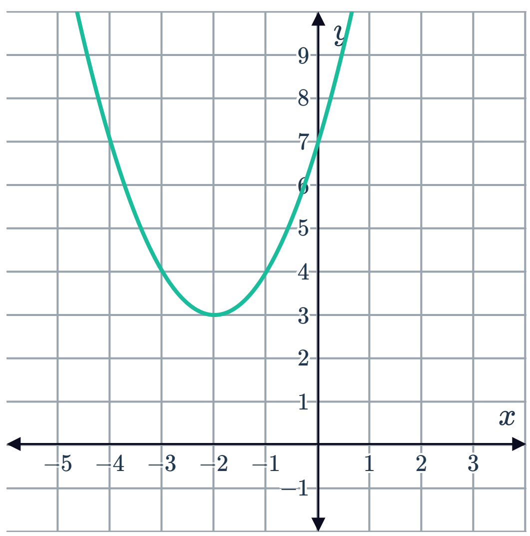
c -16

d $(4, \infty)$

e $(-\infty, 4)$

4

a



b The function has a minimum value.

c 3

d $(-2, \infty)$

e $(-\infty, -2)$

5

a No

c No

d Yes

6

a No

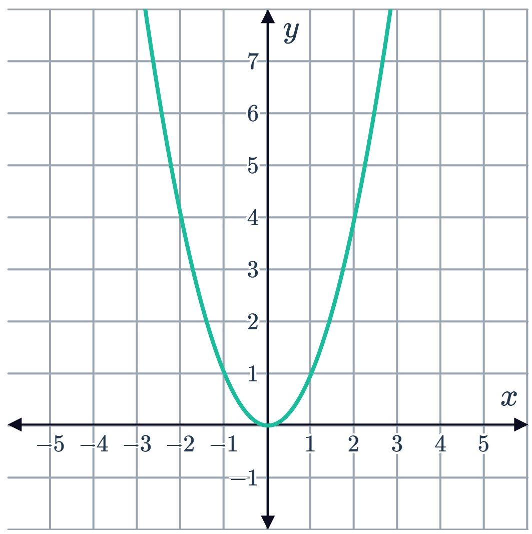
b $-\dfrac{b}{2a}$

7

a

x	-3	-2	-1	0	1	2	3
y	9	4	1	0	1	4	9

b



c No

d $x = 0$

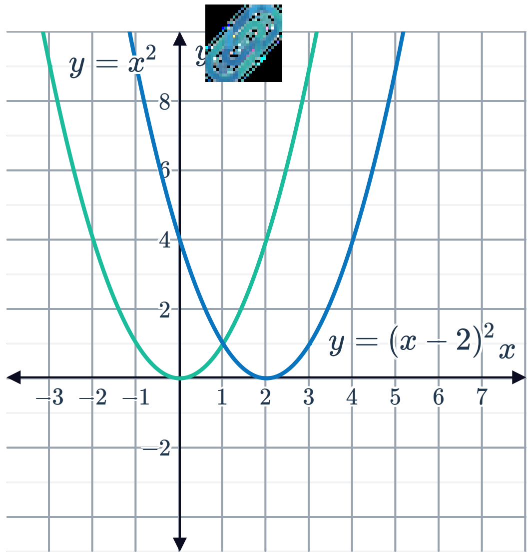
e 0

f 2

8

a Move the graph to the right by 2 units.

b

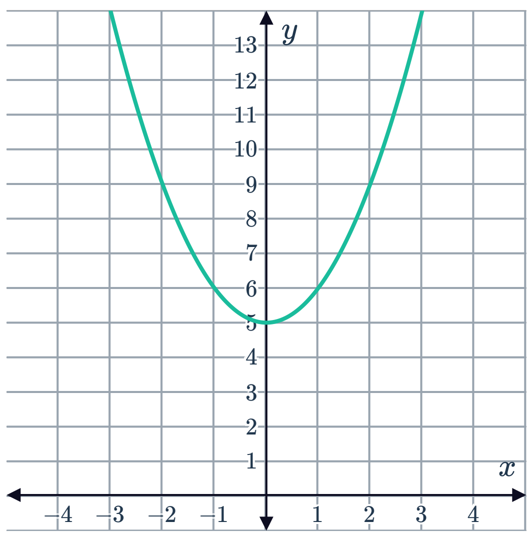


9

a

x	-3	-2	-1	0	1	2	3
y	14	9	6	5	6	9	14

b



10

a

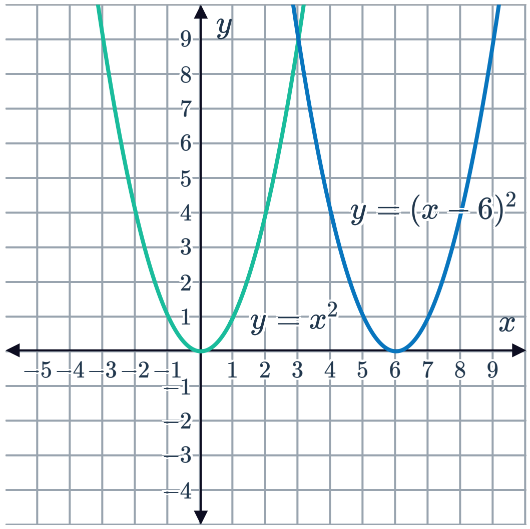
x	4	5	6	7	8
y	4	1	0	1	4

b 0

c 6

e 6 units

f



11

a

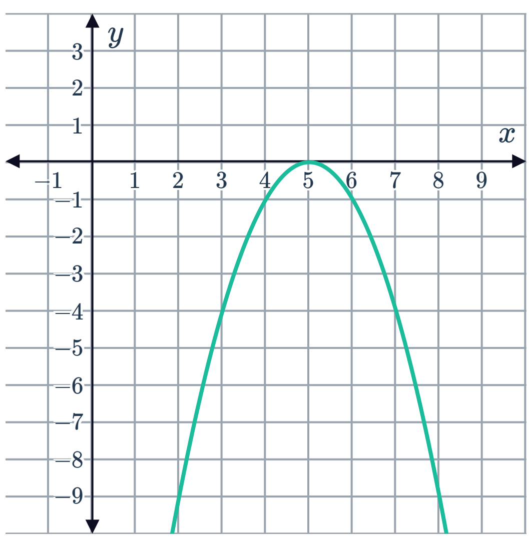
x	3	4	5	6	7
y	-4	-1	0	-1	-4

b 0

c 5

d Vertex $= (5, 0)$

e



12

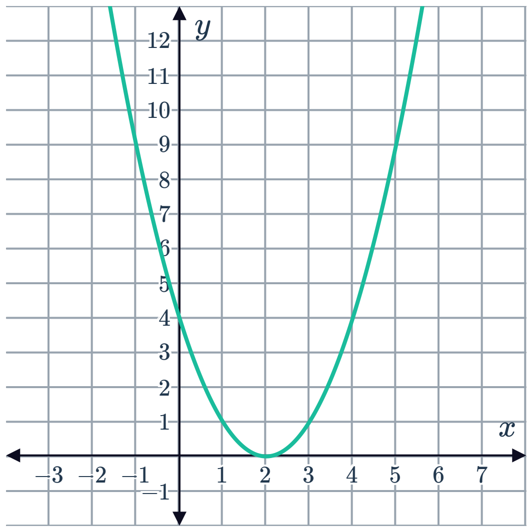
a

i $x = 2$

ii $y = 4$

iii Vertex $(2, 0)$

iv

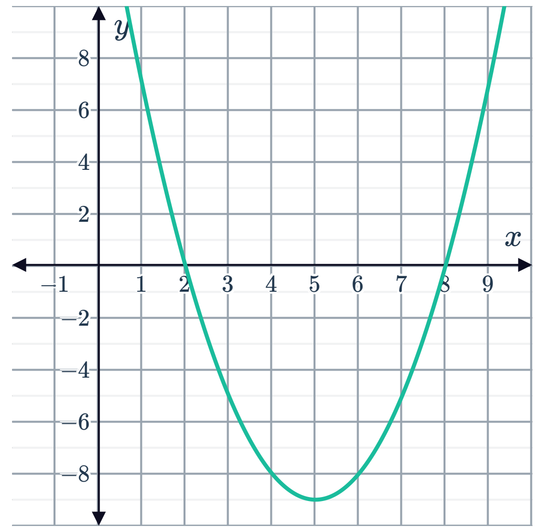


i $x = 8, x = 2$

ii 16

iii Vertex $(5, -9)$

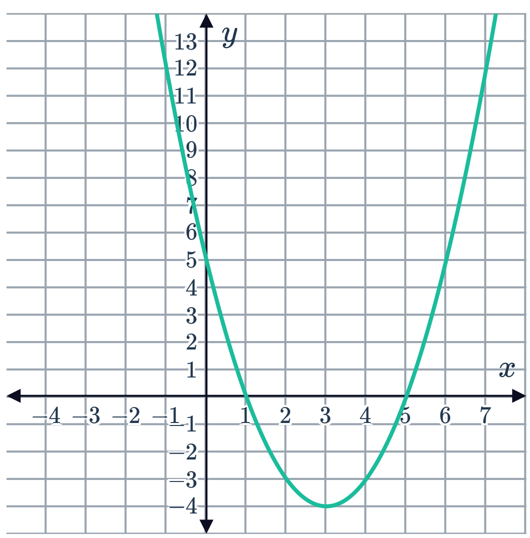
iv



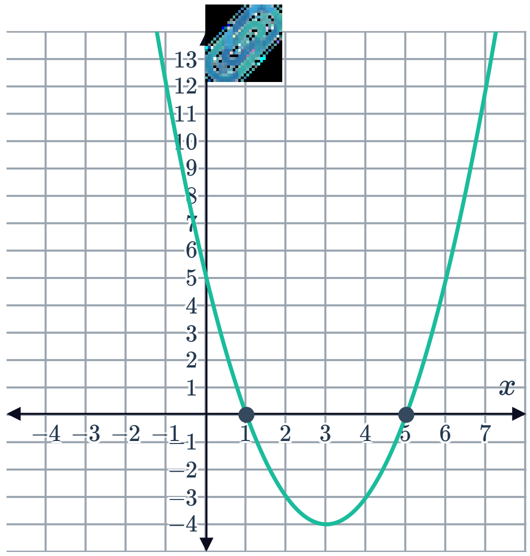
13

a $x = 1, x = 5$

b



c

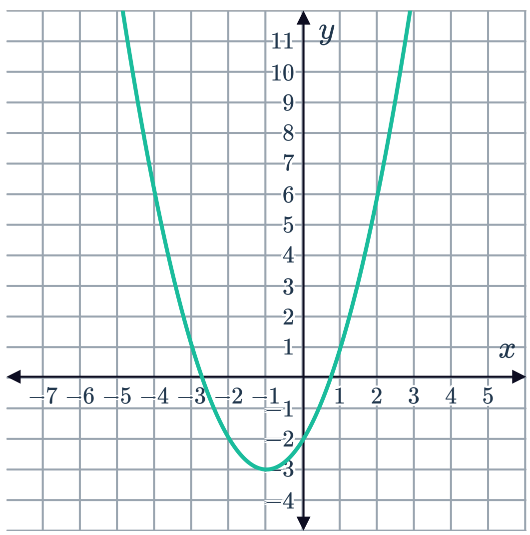


14

a $x = -1$

b -3

c



15

a $x = -3$

b -4

c $(-3, -4)$

d $(-3, -8)$

e

i Yes

ii Yes

iii No

16

a -4

b $y \geq -4$

c All real x

17

a 16

b $y \leq 16$

c All real x

18

a 0

b $y \geq 0$

c $x > 7$

19

a 0

b $y \geq 0$

c $x < -4$

20

a Minimum

b -1

c $y \geq -1$

21

a Maximum

b -2

c $y \leq -2$

22

a Minimum

b -3

c $y \geq -3$

23

a Maximum

b -3

c $y \leq -3$

24

a No

b $x = 0$

c $y = 0$